HONG KONG BAPTIST UNIVERSITY
COURSE OUTLINE

1. COURSE TITLE

Natural Language Processing and Large Language Models

2. COURSE CODE

COMP7045

3. NO. OF UNITS

3 Units

4. OFFERING DEPARTMENT

Master of Science in Data Analytics and Artificial Intelligence

5. PREREQUISITES

Basic knowledge in artificial intelligence

6. MEDIUM OF INSTRUCTION

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7. AIMS & OBJECTIVES

To give students a solid background of natural language processing and Large Language Models. To learn creative designs and implementations, including critical thinking of methodologies, problem analytics, practical techniques and tools for a wide range of downstream applications. Students after taking this course will be able to: 1) identify and creatively apply advanced techniques of natural language processing to solve downstream problems, and 2) build modules to creatively design natural language systems, and critically analyze their effectiveness; and 3) gain advanced knowledge of Large Language Models, with the ability to design, implement, and evaluate solutions for specialized tasks.

8. COURSE CONTENT

I. Introduction to Natural Language Processing (NLP) and Core Concepts

II. Text Preprocessing Techniques
- Text Preprocessing: Tokenizing, Segmentation, Lemmatization, Stemming, etc
- Regular expression

III. Statistical Language Models and Syntactic Analysis
- N-grams Language Models
- Smoothing and Evaluation
- Vector Semantics and Vector Space Model
- Part-Of-Speech Tagging and Syntactic Parsing

IV. Word Representation and Deep Neural Networks
- Word Embedding
- Recurrent Neural Networks and Convolutional Neural Networks
- Recursive Neural Networks
- Network Embedding

V. Transformers and Neural Language Models
- Attention in Sequence-to-Sequence Models
- Introduction to Transformer
- Neural Language Models

VI. Training Large Language Models (LLMs) and Foundation Models
- Pre-training and Fine-tuning
- Model Alignment, Prompting, and In-Context Learning
- Overview of Foundation Models (GPT, LLaMA, DeepSeek and others)

VII. Downstream Applications of NLP and LLMs Techniques
- Sentiment Classification
- Machine Translation
- Question Answering and Chatbots
- Retrieval Augmented Generation
- Text Summarization

9. COURSE INTENDED LEARNING OUTCOMES (CILOs)

CILO	By the end of the course, students should be able to:
CILO 1	Describe the fundamental concepts and methodologies of natural language processing and large language models
CILO 2	Explain the advantages and limitations of methods developed for different scenarios
CILO 3	Identify relevant language processing techniques to meet real-world needs
CILO 4	Apply specific methods and techniques in a wide-range of downstream applications
CILO 5	Design and evaluate the solutions to technical problems
CILO 6	Engage in life-long independent learning for problem-solving and advance state-of-the art solutions

10. TEACHING & LEARNING ACTIVITIES (TLAs)

CILO alignment	Type of TLA
1-5	Students will learn essential knowledge of natural language processing and large language models through lectures and tutorials. There will be written assignments, quizzes and final examination to evaluate the students' level of understanding.
4-5	Students will learn critical algorithms and techniques of some downstream tasks and applications through lectures and tutorials. Laboratory sessions will also be designed so that students could apply what they have learnt in lectures. There will be laboratory exercises and quizzes.
4-6	Students are required to conduct a project based on a selected topic about NLP and large language models and give a formal presentation on their proposed method. Peer evaluation by Instructor(s) and teaching assistant(s) will be involved.

11. ASSESSMENT METHODS (AMs)

Type of Assessment Methods	Weighting	CILOs to be addressed	Description of Assessment Tasks
Continuous Assessment - assignments and quizzes/tests	25 %	1-6	Assignments, quizzes and labs will be used to consolidate their knowledge and develop their skills in natural language processing and large language models.
Continuous Assessment - project	25 %	1-6	Individual/group project and report will further strength their understanding and problem solving skills. Peer evaluation will be involved.
Examination	50 %	1-6	Examination will be used to assess students' overall understanding in the concepts, algorithms, methodologies and their ability in applying these knowledge to solve problems.

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